Multiple-choice questions

|!|BCEMC0010000001|!|

Which of the following is an example of excretion in humans?

A. exhaling air from the lungs
B. coughing out sputum
C. sneezing out trapped particles from the nasal cavity
D. expelling indigestible food in faeces

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0010000003|!| ★ SI

On a field trip to a rocky shore, students observed that seaweeds grow on the upper shore (far from the sea), while no seaweed grow on the lower shore (near the sea). Several students suggested explanations for the lack of seaweed, such as:

- Seaweeds cannot grow near the sea because of the strong wave action.
- Invertebrates living on the lower shore eat all the seaweed.

The above suggested ideas are

A. observations.
B. hypotheses.
C. predictions.
D. conclusions.

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0010000004|!| ★

Which of the following characteristics of a cell indicates that it must be alive?

A. presence of a cell membrane
B. presence of genetic material
C. movement
D. respiration

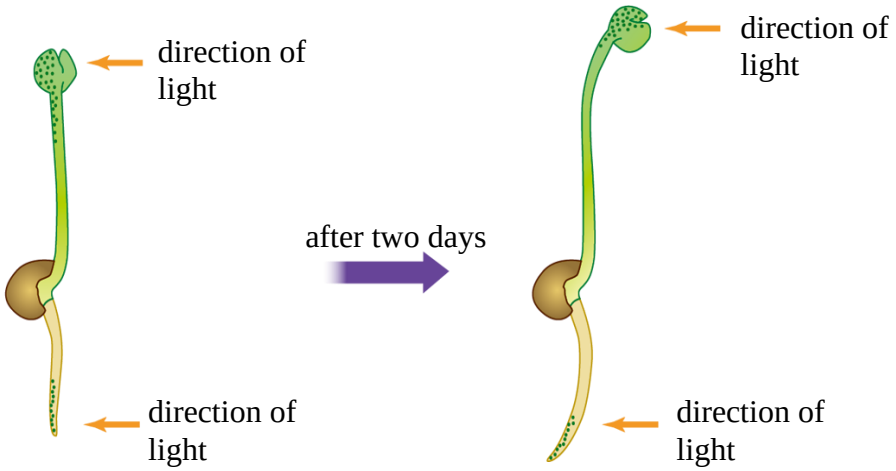
##D##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0010000005|!| ★

The diagram below shows the appearance of a seedling at the beginning of an investigation and two days later.



Which characteristic(s) of organisms is/are demonstrated in this investigation?

(1) growth
(2) movement
(3) sensitivity

A. (3) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0010000007|!| ★ SI

Susan wants to find out whether red light, instead of sunlight, can promote the growth of bean sprouts. Which of the following set-ups should be used?

	Experimental set-up	*Control set-up*
A.	Expose bean sprouts to red light	Keep bean sprouts in darkness
B.	Expose bean sprouts to red light	Expose bean sprouts to sunlight
C.	Expose bean sprouts to blue light	Expose bean sprouts to red light
D.	Keep bean sprouts in darkness	Expose bean sprouts to sunlight

##B##

Skill assessed	Mark
Practical skills	1

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|!|BCEMA0010000008|!| ★★ SI

Susan carried out an investigation to study whether red light, instead of sunlight, can promote the growth of bean sprouts. She prepared two pots of bean sprouts. Each pot contained 10 seedlings.

What is the advantage of including 10 seedlings, instead of one, in each pot?

A. to show that any difference in the results is due to the colour of light
B. to ensure that the investigation is a fair test
C. to reduce the effect of individual difference on the results
D. to study the growth of each seedling in details

##C##

Skill assessed	Mark
Practical skills	1

|!|BCEMA0010000009|!| ★★ SI

In an investigation, carbon dioxide was bubbled into a jar of water. The pH of the water was measured and recorded at 10-second intervals.

Time (s)	pH of the water
0	7.6
10	7.4
20	7.0
30	6.8
40	6.4
50	5.9

Which of the following conclusions can be drawn based on the results?

A. The pH of the water would become 3 after 120 seconds.
B. As the amount of carbon dioxide in the water increases, the water becomes more acidic.
C. As the amount of carbon dioxide in the water decreases, the water becomes more acidic.
D. The change in pH of the water is not caused by carbon dioxide.

##B##

Skill assessed	Mark
Analysing data	1

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|!|BCEMB0010000010|!| ★ **SI**

Michael wants to investigate whether the browning of cut apples is caused by exposure to light. He keeps three slices of apples in darkness for an hour. Which of the following is a suitable control set-up for his investigation?

A. Immerse three slices of apples in water
B. Keep three slices of apples in the refrigerator
C. Leave three slices of apples under a table lamp
D. Spread salad dressing on three slices of apples

##C##

Skill assessed	Mark
Practical skills	1

|!|BCEMA0010000011|!| ★★ **SI**

Michael investigated whether the browning of cut apples is caused by exposure to air. He prepared six slices of apples. He immersed three slices in water, while the other three slices were left in the air. Which of the following is/are the controlled variable(s) of his investigation?

(1) types of apple used
(2) length of time left in water or in air
(3) degree of browning of the apple slices

A. (2) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Practical skills	1

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|!|BCEMC0010000012|!|

Which of the following descriptions about the hypothesis of a scientific investigation is correct?

(1) It is a proposed explanation of an observation.
(2) It can be tested by conducting experiments.
(3) It is always true when the results of an experiment support it.

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0010000013|!| ★

Which of the following is ***not*** an example of excretion?

A. Carbon dioxide is breathed out from the lungs.
B. Sweating occurs under hot weather.
C. Urea leaves the body in urine.
D. Undigested food passes out of the body through the anus.

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0010000014|!| ★ SI

In the 18th century, smallpox was a very common disease in the UK.

A doctor called Edward Jenner discovered a way of protecting people from contracting smallpox. The following shows the process of his investigation.

- Jenner noticed that milkmaids often caught cowpox from cows.
- He also noticed that the milkmaids did not catch smallpox.
- He made a hypothesis.
- To test his hypothesis, Jenner took some pus from the blisters of a cowpox patient and scratched the pus into the skin of a healthy boy.
- The boy got infected and recovered in a few days.
- Jenner then scratched some pus from the blisters of a smallpox patient into the skin of the same boy.
- The boy did not develop smallpox.

Which of the following is the hypothesis of Jenner's investigation?

A. Cowpox was often found in milkmaids.
B. Cowpox was common in the UK.
C. All milkmaids would not develop smallpox.
D. If people were infected with cowpox, they would also become immune to smallpox.

☐

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0010000015|!| ★

Which of the following is ***not*** an example of biotechnology?

A. cloning
B. mass production of vaccines
C. development of the microscope
D. production of genetically modified food

☐

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0010000016|!|

Which of the following descriptions about the branches of biology is ***incorrect***?

	Branch of biology	***The study of***
A.	biotechnology	the use of living things or life processes to make useful products
B.	ecology	how evolution occurs
C.	physiology	how living things function
D.	taxonomy	the identification and classification of living things

☐

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0010000017|!|

Which of the following branches of biology involves the study of the interactions among organisms and their environments?

A. ecology
B. cytology
C. zoology
D. physiology

☐

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0010000018|!| ★

Which of the following is ***not*** a biological invention or discovery?

A. discovery of the structure of DNA
B. cloning of animals
C. invention of optical fibres
D. discovery of penicillin

☐

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0010000019|!|

Which of the following refers to the ability of an organism to detect changes in its environment and produce responses?

A. movement
B. irritability
C. observation
D. reproduction

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0010000020|!| ★

Which of the following statements about the common characteristics of organisms is correct?

A. Nutrition is the breakdown of food to release energy.
B. Defaecation is an example of excretion.
C. The growing of plants towards light is an example of sensitivity.
D. The rapid inflation of a puffer fish upon stimuli is an example of growth.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0010000021|!|

In animals, metabolic wastes are excreted in

(1) sweat.
(2) urine.
(3) exhaled gas.

A. (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0010000022|!|

Which of the following statements about respiration is correct?

A. Respiration refers to the breathing in and out of oxygen and carbon dioxide.
B. Plants carry out photosynthesis instead of respiration.
C. Organisms obtain energy by breaking down food in their cells through respiration.
D. Organisms can carry out respiration only when oxygen is available.

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0010000023|!|

Movement, growth and repair of body tissues require energy. Which of the following processes provides energy for these activities?

A. photosynthesis
B. excretion
C. respiration
D. active transport

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0010000024|!|

Which of the following are the characteristics found in all living organisms?

(1) movement from place to place
(2) excretion
(3) nutrition
(4) irritability

A. (1) and (2) only
B. (2) and (3) only
C. (2), (3) and (4) only
D. (1), (2), (3) and (4)

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0010000025|!| ★

Which of the following combinations of the characteristic of organisms and its example is ***incorrect***?

	Characteristic	***Example***
A.	Reproduction	A cow gives birth to a calf.
B.	Sensitivity	A honeybee flies out from the honeycomb.
C.	Respiration	A green plant produces and releases carbon dioxide at night.
D.	Nutrition	A tapeworm obtains nutrients from the human intestine.

☐

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0010000026|!| ★

Which of the following processes commonly found in plants require(s) the exchange of gases with the surroundings?

(1) respiration
(2) nutrition
(3) growth

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3)

☐

##B##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0010000027|!| ★

Which of the following common characteristics of organisms enables an animal to escape from predators?

A. excretion
B. sensitivity
C. reproduction
D. nutrition

☐

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMC0010000028|!|

Which of the following is ***not*** an example of excretion?

A. Plants store waste products in leaves which are shed off.
B. Animals release carbon dioxide in exhaled gas.
C. Animals discharge urine.
D. Animals expel faeces from their bodies.

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0010000029|!|

Which of the following is ***not*** a common characteristic of organisms?

A. breathing
B. sensitivity
C. movement
D. reproduction

##A##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0010000030|!|

Which of the following processes remove the wastes produced by chemical reactions from our bodies?

(1) sweating
(2) urination
(3) egestion
(4) breathing

A. (1), (2) and (3) only
B. (1), (2) and (4) only
C. (2), (3) and (4) only
D. (1), (2), (3) and (4)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMB0010000031|!| ★

Which of the following organisms can carry out nutrition, respiration, excretion and movement?

(1) bacteria
(2) jelly fish
(3) sunflower

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0010000032|!|

Which of the following is ***not*** a characteristic of all living organisms?

A. nutrition
B. exhalation
C. excretion
D. irritability

##B##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0010000033|!|

Which of the following processes remove metabolic wastes from the body of an organism?

A. respiration
B. irritability
C. excretion
D. movement

##C##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0010000034|!|

Which of the following processes produces heat to keep the human body temperature stable?

A. nutrition
B. reproduction
C. respiration
D. growth

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0010000035|!| ★

Which of the following organs is ***not*** involved in excretion?

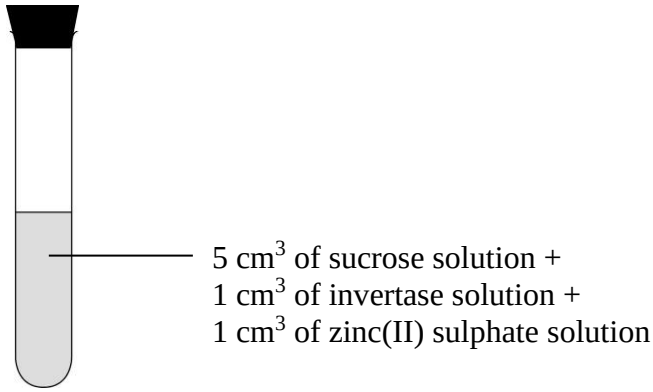
A. lungs
B. skin
C. kidneys
D. rectum

##D##

Skill assessed	Mark
Applying concepts	1

|!|BCEMB0010000036|!| ★ SI

Peter carried out an investigation to study the effect of zinc(II) ions on the activity of invertase (an enzyme that catalyses the breakdown of sucrose into glucose and fructose). He prepared the experimental set-up by adding the following solutions into a test tube, as shown below:



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Which of the following is/are the correct control set-up(s) in this investigation?

(1) 5 cm^3 of sucrose solution + 2 cm^3 of distilled water
(2) 5 cm^3 of sucrose solution + 1 cm^3 of invertase solution + 1 cm^3 of distilled water
(3) 5 cm^3 of sucrose solution + 1 cm^3 of invertase solution + 1 cm^3 of zinc(II) nitrate solution

A. (2) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only

##A##

Skill assessed	Mark
Practical skills	1

|!|BCEMC0010000037|!|
What is a hypothesis in scientific investigation?
A. an idea that cannot be tested by experiments
B. an experiment that is used to test a theory
C. a proposed explanation of observations that can be tested
D. an explanation of observations that is proved wrong after experiments

##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMC0010000038|!|
The following shows the steps in scientific investigations. Arrange them in the correct order.

(1) making observation and asking questions about the observation
(2) drawing a conclusion
(3) carrying out experiments
(4) making a hypothesis
(5) collecting data

A. (1) → (2) → (3) → (4) → (5)
B. (1) → (4) → (2) → (5) → (3)
C. (1) → (4) → (3) → (5) → (2)
D. (4) → (3) → (1) → (5) → (2)

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##C##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMB0010000040|!| ★ SI

In the 1840s, many women died from infections after giving birth in hospitals. A doctor called Semmelweiss found that the death rate of mothers was higher in ward X than in ward Y.

Ward X is where medical students dissected corpses and delivered babies. Ward Y is where midwives delivered babies.

Which of the following is a testable hypothesis for the higher death rate in ward X?

A. Midwives are more skilful in delivering babies than medical students.
B. The higher death rate occurred by chance.
C. Medical students transferred germs from corpses to the mothers.
D. The women who gave birth in ward X were poorer than those in ward Y.

##C##

Skill assessed	Mark
Applying concepts	1

|!|BCEMC0010000041|!|

Which of the following laboratory rules are correct?

(1) Pay attention to hazard warning labels on reagent bottles.
(2) Wash your hands before eating in the laboratory.
(3) Wear disposable gloves when handling living animals.
(4) Sterilize dissecting instruments after use.

A. (3) and (4) only
B. (1), (3) and (4) only
C. (2), (3) and (4) only
D. (1), (2), (3) and (4)

##B##

Skill assessed	Mark
Understanding basic concepts	1

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|!|BCEMC0010000042|!|

Which of the following is correct when the result of an experiment does ***not*** match the prediction made based on a hypothesis?

A. There is no conclusion.
B. Assumptions have to be made.
C. The investigation does not require a hypothesis.
D. The hypothesis has to be revised.

##D##

Skill assessed	Mark
Understanding basic concepts	1

|!|BCEMA0010000043|!| ★★ SI

Directions: The following two questions refer to the following investigation.

Peter wants to study competition between two types of seedlings X and Y. He prepared four pots of soil with the same mass. He planted different numbers of seeds in each pot, as shown in the table below.

Pot	Number of seeds planted	
	X	Y
1	5	5
2	10	0
3	0	10
4	10	10

Table 1

Peter put the pots on a window sill for four weeks.
Each morning, he gave each pot 20 cm^3 of water.
After four weeks, he removed the seedlings from the pots and weighed the seedlings.

Which of the following is an independent variable in the above investigation?

A. the mass of soil in each pot
B. the amount of water received by the seedlings
C. the number of seeds planted
D. the weight of the seedlings after four weeks

##C##

Skill assessed	Mark
Practical skills	1

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|!|BCEMA0010000044|!| ★★ SI

Table 2 below shows the results.

Pot	Total mass of seedlings after four weeks (g)	
	X	Y
1	9	12
2	29	0
3	0	33
4	22	28

Table 2

Which of the following conclusions can be drawn based on the results?

A. Seedlings X and Y compete for water and nutrients in the soil.
B. The higher the number of seedlings per pot, the smaller the effect of competition.
C. Competition favours the growth of seedling Y.
D. Competition adversely affects the growth of seedlings X and Y.

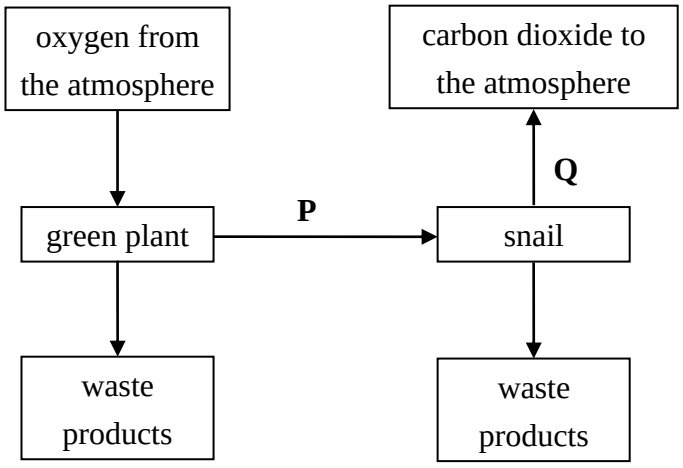
□

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0010000045|!| ★

Directions: The following two questions refer to the diagram below, which outlines some processes carried out by green plants and snails.



Which characteristic of organisms is represented by arrow P?

A. reproduction
B. sensitivity
C. nutrition
D. excretion

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##C##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0010000046|!| ★

Which characteristic of organisms is represented by arrow Q?

A. respiration
B. sensitivity
C. movement
D. excretion

##D##

Skill assessed	Mark
Analysing data	1

|!|BCEMB0010000047|!| ★ SI

Directions: The following two questions refer to the passage below, which is about an investigation of the pathogen that causes a new disease:

> A new illness, called disease M, had arisen. Patients of disease M showed symptoms similar to those of disease P. The same drugs can be used to treat diseases M and P. A scientist suspected that diseases M and P were caused by the same pathogen. The scientist analysed the DNA fingerprints of the pathogens M and P. She found that their DNA fingerprints differ. Another hypothesis and tests were set.

Which of the following is the hypothesis of the investigation?

A. Disease M showed symptoms similar to those of disease P.
B. Diseases M and P were caused by the same pathogen.
C. The scientist analysed the DNA fingerprints of the pathogens M and P.
D. The scientist found that the DNA fingerprints of the pathogens M and P differ.

##B##

Skill assessed	Mark
Applying concepts	1

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|!|BCEMB0010000048|!| ★ **SI**

Which conclusion can be drawn from this investigation?

A. Diseases M and P were caused by the same pathogen.

B. Diseases M and P were caused by different pathogens.

C. Mutation occurs in the pathogen that causes disease M.

D. The drug which can be used to treat disease P could treat disease M.

□

##B##

Skill assessed	Mark
Applying concepts	1

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